



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)

Department of Mathematics

Undergraduate Program

COURSE PLAN	SEMESTER: Fall 2024-2025
I. Course Core and Title MAT 2202 Matrices, Vectors and Fourier Analysis II. Credit 3 credit hours (3 hours of theory per week) III. Nature Core Course for CS and Engineering IV. Prerequisite MAT 2201: Complex Variables, Laplace, and Z-Transformations MAT 1205: Integral Calculus & Ordinary Differential Equation	V. Vision: Our vision is to be the preeminent Department of Mathematics through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context. VI. Mission: The mission of the Department of Mathematics of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description:

- Discussion about Matrix Algebra, inverse of a matrix, row echelon form, reduced row echelon form
- Solution of the system of linear equations, application of system of linear equations
- Discussion about periodic function, Fourier series of a periodic function and half range Fourier series, Fourier integral, Fourier sine and cosine integrals
- Fourier transformation, inverse Fourier transformations, and discrete Fourier transform
- Discussion about eigenvalues, eigen vectors and Cayley-Hamilton theorem
- Discussion on Linear combination and linear dependency of vectors
- Discussion about coordinate systems (Rectangular, Cylindrical & Spherical) and conversion to each other, Coordinate transform of vectors and points
- Discussion about differentiation of a vector function, gradient, and directional derivative of a scalar function
- Discussion about Divergence theorem and Stokes theorem in Cartesian, Cylindrical and Spherical co-ordinate systems

VIII – Course outcomes (CO) Matrix:

By the end of this course, students should be able to:

COs*	CO Description	Level of Domain ***			PO Assessed ****
		C	P	A	
CO1**	<i>Predict and determine</i> the fundamental concepts of linear algebra, co-ordinate systems, vector calculus and Fourier analysis.	3			PO-a-2
CO2	<i>Solve</i> the system of linear equations, ordinary differential equations with the help of gained fundamental concepts related to linear algebra. <i>Demonstrate</i> Fourier series, Fourier transform and Fourier integral of function.	3			PO-a-2

CO3 **	Solve the real-life problem such as networking problem, circuit problem and cryptography, optimization problem with the help of gained fundamental concepts related to linear algebra.	3			PO-a-4
CO4	Apply the theory of vector calculus in different geometrical surfaces (open/closed surface).	3			PO-a-2

C: Cognitive; P: Psychomotor; A: Affective Domain

* CO assessment method and rubric of COs assessment is provided in later section

** COs will be mapped with the Program Outcomes (POs) for PO attainment

*** The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.

**** The numbers under 'PO Assessed' column represent the POs each CO corresponds to.

IX – Topics to be covered in the class and/or lab*:

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1	Matrix algebra, inverse of a matrix. Elementary row transformation, row echelon form, reduced row echelon form.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 2	CO2	Solution of system of linear equations by using Gaussian elimination method, matrix inverse method and determinant (Cramer's rule).	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 3	CO3	Applications of matrices (Networking Problems, Circuit problems, Robotics & Cryptography) in real life.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance, Quiz-1
Week 4	CO3	Linear programming problems (Maximization/ Minimization or Optimization).	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 5	CO1, CO2	Fourier Analysis: Periodic functions and periodicity of a given function, even and odd function, Fourier series in real form, odd and even extensions.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance, Quiz-2
Week 6	CO1, CO2	Fourier series in complex form, Half range Fourier sine and cosine series, Fourier Integral.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance, Quiz-3
Week 7	CO1, CO2	Finite and infinite Fourier transform and Discrete Fourier Transform.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Midterm (Week 7)				
Week 8	CO1, CO2	Eigenvalues & eigenvectors; applications of eigenvalues & eigenvectors to solve the system of ordinary differential equations (ODEs).	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 9	CO1	Cayley-Hamilton theorem and it's applications, vector space. Coordinate systems (cartesian coordinate, cylindrical coordinate, and spherical coordinate).	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance, Quiz-1

Week 10	CO1	Gradient, directional derivative and Laplacian of scalar function.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 11	CO1, CO4	Divergence and curl of a vector function, Gauss's Divergence theorem in cartesian coordinate.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance, Quiz-2
Week 12	CO1, CO4	Gauss's Divergence theorem in cylindrical and spherical coordinates.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 13	CO1, CO4	Stokes theorem in cartesian and cylindrical coordinate system.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance
Week 14	CO1, CO4	Stokes Theorem in spherical coordinate system.	Lecture, Brainstorming, Problem solving, Question-answer.	Class Performance, Quiz-3
Final term (Week 15)				

* The faculty reserves the right to change, amend, add, or delete any of the contents.

X – Mapping of PO to Courses and K, P, A

PO Indicator ID	PO Indicators Definition (As per the requirement of WKs)	Domain	K	P	A
PO-a-2	Apply information and concepts of mathematics with the familiarity of issues.	Cognitive Level 3 (Applying)	K2		
PO-a-4	Apply information and concepts in <i>specialized engineering sciences</i> with the in-depth of analysis of a complex computer science and engineering problem.	Cognitive Level 3 (Applying)	K4	P1 P3 P7	

XI – K, P, A Definitions

Indicator	Title	Description
K2	Conceptual based mathematics	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K4	Forefront specialist knowledge for practice	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline

Indicator	Title	Description
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P3	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
P7	Interdependence	Are high level problems including many component parts or sub-problems

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	<i>Predict and determine</i> the fundamental concepts of linear algebra, co-ordinate systems, vector calculus and Fourier analysis.	PO-a-2	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment
CO2	<i>Solve</i> the system of linear equations, ordinary differential equations with the help of gained fundamental concepts related to linear algebra. <i>Demonstrate</i> Fourier series and Fourier transform and Fourier integral of function.	PO-a-2	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment
CO3	<i>Solve</i> the real-life problem such as networking problem, circuit problem and cryptography, optimization problem with the help of gained fundamental concepts related to linear algebra.	PO-a-4	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment
CO4	<i>Apply</i> the theory of vector calculus in different geometrical surfaces (open/closed surface).	PO-a-2	Quiz/Term Question & Assignment	Rubric for Quiz/Term Question & Assignment

XIII – Evaluation and Assessment Criteria

CO1: <i>Predict and determine</i> the fundamental concepts of linear algebra, co-ordinate systems, vector calculus and Fourier analysis.				
Assessment Attribute/Criteria	Missing/Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Definition	Did not attend or wrong answer	Concept is not clear	Explanation is clear but not thorough	Explained the concept thoroughly
Mathematical formation	Answered wrong mathematical form or did not attend	Mathematical form is partially correct	Mathematical form is correct but need some more modification.	Written mathematical form correctly
Example(s)	Did not attend or wrong answer	Example is not appropriate	Given the example but concept is not discussed clearly	Given example(s) of fundamental concept, theory, operations and discussed clearly
Correctness	Did not attend or wrong answer	Answer is partially correct	Arrived at correct answer but some steps are not shown	Arrived at correct answer with no error showing every step of calculation

CO2: Solve the system of linear equations, ordinary differential equations with the help of gained fundamental concepts related to linear algebra. *Demonstrate* Fourier series and Fourier transform and Fourier integral of function.

Assessment Attribute/Criteria	Missing/Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Formulation	No formulation is given or wrong formulation	Did not accurately present the problem	Accurately present the problem but some logic is not clearly present	Written the formulae for respective problem correctly with logic
Selection of methodology	Did not attend or selected wrong method	Correct method chosen but computation process remains incomplete	Problem explained with the correct method but a few small mistakes	Classified the problem with the relevant method correctly
Fulfil the requirement	Did not attend or did not fulfill any requirements	Makes some progress in applying the method but errors in calculations or steps	Applies the correct method with some minor errors or omissions in calculations	Solved the problem by showing necessary steps with all calculation
Correctness of Answer	Did not attend or totally wrong answer	Answer is partially correct	Arrived at correct answer but some steps are not shown	Arrived at correct answer with no error showing every step of calculation

CO3: Solve the real-life problem such as networking problem, circuit problem and cryptography, optimization problem with the help of gained fundamental concepts related to linear algebra.

Assessment Attribute/Criteria	Missing/Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Construct mathematical model	No formulation is given or wrong mathematical model	Did not accurately Construct the problem in mathematical model	Accurately present the problem in appropriate mathematical model but some logic is not clearly present	Construct mathematical model based on the problems
Selection of methodology	Did not attend or selected wrong method	Correct method chosen but computation process remains incomplete	Problem explained with the correct method but a few small mistakes	Classify the problem with the relevant method
Fulfil the requirement	Did not attend or did not fulfill any requirements	Makes some progress in applying the method but errors in calculations or steps	Applies the correct method with some minor errors or omissions in calculations	Solve the problem by showing necessary steps with calculation
Correctness of Answer	Did not attend or totally wrong answer	Answer is partially correct	Arrived at correct answer but some steps are not shown	Arrived at correct answer with no error showing every step of calculation

CO4: Apply the theory of vector calculus in different geometrical surfaces (open/closed surface).				
Assessment Criteria	Missing/Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Selection of Methodology	No formulation is given or wrong mathematical model	Correct method chosen but computation process remains incomplete	Problem explained with the correct method but a few small mistakes	Identify correct theory for relevant problem(s)
Calculation	Answered wrong mathematical form or did not attend	Makes some progress in applying the method but errors in calculations or steps	Applies the correct method with some minor errors or omissions in calculations	Provide step by step calculation to evaluate surface, volume and line integral for respective geometrical surfaces
Justify the theory	Did not attend or wrong answer	Make some progress in verifying the theory but some logic present partially	Verify the theory but did not show necessary steps	Verify the theory or results
Correctness of answer	Did not attend or totally wrong answer	Answer is partially correct	Arrived at correct answer but some steps are not shown	Arrived at correct answer with no error showing every step of calculation

XIV- Course Requirements

- Students are expected to attend at least 80% of the class.
- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.

XV – Evaluation & Grading System*

Midterm and Final term	
Quiz	40%
Class Attendance	10%
Class Performance/Assignment	10%
Mid/ Final term Examination	40%
Total	100%
Final Grade/ Grand Total	
Midterm:	40%
Final Term:	60%

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

* The evaluation system will be strictly followed as per the AIUB grading policy.

* CO attainment will be achieved with 60% of the evaluation marks.

XVI – Textbook/ References

1. Elementary Linear Algebra: Applications Version-H. Anton and C. Rorres, 11th edition, Wiley, 2013.
2. Fundamentals of Applied Electromagnetics-Fawwaz T. Ulaby and Umberto Ravaioli, 7th edition, Pearson, 1999.
3. An Introduction to Mathematical Cryptography-Jeffrey Hoffstrein, Jil Pipher and Joseph H. Silverman; Publisher's: Springer.
4. Linear Programming-Thomas Ferguson
5. Linear Algebra and Its Application-David C. Lay and Steven R Lay, 5th Edition, Pearson, 1997.
6. Advance Engineering Mathematics-E. Kreyszig, 10th edition, John Wiley and Sons, 2010.
7. Lecture notes

XVII - List of Faculties Teaching the Course

FACULTY NAME	FACULTY NAME
1. Dr. Kh. Abdul Maleque	11. Md. Hassan Faruk
2. Dr. Mohammed Jashim Uddin	12. Shakil Mojumdar
3. Dr. Madhabi Islam	13. Asif Chowdhury
4. Tanzia Zerin Khan	14. Kazi Habib Ullah
5. Prodip Kumar Ghose	15. Mansura Akter Mome
6. Ayesha Siddiqua	16. Md. Majibul Hasan Imran
7. Dr. Mahadeb Kumar Das	17. Umma Kulsum
8. Dr. Dilruba Yasmin	18. Sadia Shihab Sinje
9. Dr. Salaika Parvin	19. Afsana Akter Mitu
10. Md. Mehedi Hasan	

XVIII – Verification

Prepared by: ----- Tanzia Zerín Khan <i>Course Convener</i> Date:.....	Moderated by: ----- Dr. M. Mahmudul Hasan <i>Point Of Contact</i> <i>OBE Implementation Committee</i> Date:.....	Checked by: ----- Dr. M. Mostafizur Rahman <i>Head, Department of Mathematics</i> Date:.....
Verified by: Dr. Md. Abdullah-Al-Jubair <i>Director</i> <i>Faculty of Science & Information Technology</i> Date:.....	Certified by: Prof. Dr. Dip Nandi <i>Associate Dean,</i> <i>Faculty of Science & Information Technology</i> Date:.....	Approved by: Mr. Mashioor Rahman <i>Dean,</i> <i>Faculty of Science & Information Technology</i> Date:.....